RC CAR FIELD TEST AND TUNING PLAN

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This document defines the recommended testing and tuning order now that manual control is working. The goal is to validate the car safely, tune the control stack in the right order, and avoid wasting time tuning autonomy on top of unverified base behavior.

1. CURRENT STATUS

The current baseline is:

- manual throttle works
- manual steering works in principle
- joystick input works
- LiDAR connects
- webcam starts
- the controller software starts and runs

Known hardware limitation:

- the steering servo is still jittery until the steering PWM path is improved, likely with a PCA9685 or another more stable servo-driving solution

2. TEST PHILOSOPHY

The tuning order matters.

Do not start by tuning sidewalk autonomy on an unstable drive base. First confirm that the car drives correctly by hand, the sensors produce believable data, and the control outputs are physically correct. Only then should autonomy tuning begin.

The correct sequence is:

1. Validate the base vehicle
2. Tune manual-control fundamentals
3. Validate sensors while manually driving
4. Run low-speed autonomy tests in a closed course
5. Tune sidewalk-following behavior from logs

3. PHASE 1: BASE VEHICLE VALIDATION

The purpose of this phase is to verify that the physical car behaves correctly before any autonomy logic is trusted.

3.1 STEERING

Verify:

- steering center is correct when steer is zero
- full-left and full-right do not mechanically bind
- the steering direction matches controller input
- the servo holds a commanded position consistently

If this fails, tune:

- steering center trim
- steering sign/direction
- max steering range
- servo power and PWM stability

3.2 DRIVETRAIN

Verify:

- forward really moves the car forward
- reverse really moves the car backward
- braking actually stops motion
- the car rolls and coasts in a predictable way
- gear changes behave correctly

If this fails, tune:

- motor polarity and direction mapping
- acceleration ramp
- brake rate
- coasting rate
- reverse speed limit

3.3 SAFETY

Verify:

- releasing controls returns the car to a safe state
- manual input can always override autonomy
- emergency stop behavior is immediate enough

If this fails, stop tuning and fix the base control path first.

4. PHASE 2: MANUAL-CONTROL TUNING

The purpose of this phase is to make the car easy to drive manually and physically stable.

Tune in this order:

1. steering center
2. steering limit
3. acceleration feel
4. brake feel
5. low-speed precision

Desired result:

- the car tracks straight when commanded straight
- small steering inputs produce smooth path changes
- throttle is predictable at low speed
- braking is consistent and does not feel delayed

5. PHASE 3: SENSOR VALIDATION WHILE DRIVING MANUALLY

This phase checks whether the autonomy sensors are actually good enough under real motion.

5.1 HALL SENSOR

Verify:

- pulses increase while the car is moving
- reported speed changes with actual speed
- the speed estimate drops when the car stops

If this fails, tune:

- pulse interpretation
- wheel circumference constant
- pulses-per-revolution constant
- filtering and smoothing

5.2 FRONT ULTRASONIC

Verify:

- reported distances are believable
- close objects are detected consistently
- readings do not randomly jump under normal conditions

If this fails, tune:

• sensor mounting
• trigger/echo timing handling
• stop distance threshold

5.3 LIDAR

Verify:

• LiDAR remains connected while driving
• scan quality stays stable when the car moves
• heading selection does not jump erratically
• nearby obstacles show up consistently

If this fails, tune:

• LiDAR mounting stability
• forward angle range
• heading window size
• stop distance and clearance thresholds

5.4 WEBCAM

Verify:

• the camera sees enough of the sidewalk ahead
• the camera still sees the path in both sun and shade
• boundary confidence does not collapse too often
• image vibration does not make heading estimates unusable

If this fails, tune:

• camera height
• camera tilt
• field of view usage
• image preprocessing thresholds
• confidence thresholds

6. PHASE 4: CLOSED-COURSE AUTONOMY TEST

Do not start on a public sidewalk.

Build or choose a private sidewalk-like test route first, including:

• a straight section
• a gentle turn
• a sharper turn
• a shaded section
• a partial obstacle
• a fully blocked section

Start autonomy at very low speed.

Initial autonomy goals:

• stay on the drivable path
• stop safely when uncertain
• avoid sudden steering oscillation
• produce usable logs for later tuning

7. PHASE 5: AUTONOMY TUNING ORDER

Tune the following in order.

7.1 STEERING AND HEADING

Tune:

- `MAX_TARGET_HEADING_DEG`
- camera steering gain
- LiDAR heading influence

Watch for:

- oversteer
- understeer
- left-right oscillation
- failure to commit to turns

7.2 SPEED POLICY

Tune:

- `AUTONOMOUS_CAUTIOUS_PWM`
- `AUTONOMOUS_TURN_PWM`
- `AUTONOMOUS_CRUISE_PWM`

Watch for:

- entering turns too fast
- failing to move when it should
- braking too late
- driving too aggressively when confidence is low

7.3 OBSTACLE AND STOP LOGIC

Tune:

- `FRONT_FAILSAFE_STOP_CM`
- `FORWARD_OBSTACLE_STOP_DISTANCE_M`
- `PARTIAL_BLOCKAGE_MIN_CLEARANCE_M`

Watch for:

- stopping too early
- stopping too late
- attempting to pass gaps that are too tight
- failing to stop when the path is truly blocked

7.4 BOUNDARY LOSS AND RECOVERY

Tune:

- `LOW_CAMERA_CONFIDENCE`
- `HIGH_CAMERA_CONFIDENCE`
- driveway-cut detection timing

Watch for:

- stopping too often in harmless shade
- continuing too confidently after losing boundaries
- poor recovery after brief uncertainty

8. PASS/FAIL CHECKLIST

The car should pass these before serious sidewalk autonomy testing.

PASS CRITERIA

- tracks straight in manual mode
- steering direction is correct
- throttle and brake are predictable

• hall sensor reports movement correctly
• front ultrasonic reads believable distances
• LiDAR remains connected during motion
• camera remains usable in motion
• autonomy can be disengaged instantly with manual input

FAIL CONDITIONS

• servo jitters enough to change path unpredictably
• steering center drifts
• throttle or brake mapping is inconsistent
• LiDAR disconnects while moving
• camera confidence collapses constantly
• car oscillates rapidly left-right
• stop behavior is too late or too weak

9. WHAT TO RECORD FROM EACH TEST RUN

For each field run, record:

• route type
• lighting condition
• speed used
• whether the car stayed on path
• whether it oscillated
• whether it stopped unnecessarily
• whether it missed a turn
• whether it handled shade correctly
• whether it needed manual intervention

This is more useful than vague impressions.

10. RECOMMENDED IMMEDIATE NEXT STEPS

The recommended order from here is:

1. finish the steering PWM fix with a PCA9685 or equally stable servo solution
2. verify steering center and steering limits
3. verify hall sensor and front ultrasonic with the car moving
4. run a manual sensor-logging test in sun and shade
5. begin a very slow closed-course autonomy test

11. MOST IMPORTANT RULE

The first real autonomy milestone is not speed, smoothness, or fancy obstacle behavior.

The first real milestone is:

The car does not leave the sidewalk-like drivable path.

That should stay the top criterion until the platform is physically stable and sensor confidence is trustworthy.